Shaun Sekura

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EDUCATION

The Ohio State University — Columbus, OH
B.S Mechanical Engineering, Robotics and Autonomous Systems Minor — Expected December 2027
Recent Coursework: Fluids, Control Systems, Sensors, Machine Elements, Kinematics, System/Vibrational Dynamics, Circuits, Thermodynamics, Machine Design, Numerical Methods, Mechanics of Materials, Linear Algebra/Diff Eq

TECHNICAL SKILLS

- **Software:** SolidWorks, CATIA, Ansys, Abaqus FEA, Siemens Test Lab, Python, MATLAB, Excel, Arduino IDE
- **Lab Research/Processes:** CAD Software, FEA Analysis, Machining, Equipment Operation, Modal Testing
- **Fabrication:** CNC Machining, Vertical Mill, Lathe, 3D Printing, Soldering/Wiring (Arduino), Drafting/GD&T

EXPERIENCE

Mechanical Engineering Intern — Akron, OH
Hankook Tire — Summer 2026
- Developed Python and Excel pipeline to curve-fit stress-strain data across 244 material datasets, computing tangent moduli via numerical differentiation and achieving 93% curve-fit accuracy through least-squares error analysis.
- Validated a General Motors wheel through nonlinear FEA simulation in Abaqus and experimental modal analysis in Siemens Test Lab, correlating virtual and physical results to assess structural performance.
- Supported the Physical-to-Virtual initiative through testing and validation tasks aimed at improving virtual-to-physical test correlation for automotive clients including Rivian, Tesla, GM, Lucid, Mercedes Benz, etc.

Undergraduate Teaching Assistant — The Ohio State University
Intro To Mechanical Design (MECHENG 2900) — Spring 2026 – Present
- Mentored 244 students/semester in circuit construction and embedded systems programming.
- Communicated complex engineering concepts clearly to students with varying technical backgrounds.

System Design & Integration Engineer — Center For Automotive Research
EcoCAR Competition Team — Fall 2025 – Present
- Designed and analyzed 10+ vehicle components using SOLIDWORKS simulations to test and improve strength-to-weight ratios and general vehicle design.
- Collaborated across sub-teams (electronics, efficiency, automation, etc.) to integrate structural designs into final build for testing.

Formula SAE — Center For Automotive Research
Student Engineer — Fall 2024 – Spring 2025
- Assisted with mechanical design, fabrication, and assembly of a Formula SAE race car.
- Participated in vehicle testing and iterative improvements to support performance and reliability.

American Society of Mechanical Engineers (ASME) — The Ohio State University
Executive Board — Fall 2024 – Spring 2026
- Organized social events and design challenges, increasing club participation by 320% compared to previous years.
- Coordinated industry speaker series and professional networking events for 100+ members.

PROJECTS — SHAUNSEKURA.COM

Sustainable Rocket Redesign
- Used Python (AeroSandbox library) to build a reduced order simulation of the rocket's descent.
- Used SOLIDWORKS to reverse engineer a simplified model of the top of Starship.

Reverse Engineering of Motor Mount
- Reverse engineered a Cadillac Lyriq motor mount and worked with team to reinstall the part into the car.
- Used CATIA V5 to design mount and performed FEA Analysis to determine stress on mount.

INTERESTS

- Hiking, Music, Tennis, Basketball, Strategy Games, Competitive Trivia, Sushi